

# Experiment - 01

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(1.)

Given a table of candidates and their skills, you're tasked with finding the candidates best suited for an open Data Science job. You want to find candidates who are proficient in Python, Tableau, and PostgreSQL.

Write a query to list the candidates who possess all of the required skills for the job. Sort the output by candidate ID in ascending order.

## Assumption:

- There are no duplicates in the `candidates` table.

## `candidates` Table:

| Column Name  | Type    |
|--------------|---------|
| candidate_id | integer |
| skill        | varchar |

## `candidates` Example Input:

| candidate_id | skill      |
|--------------|------------|
| 123          | Python     |
| 123          | Tableau    |
| 123          | PostgreSQL |
| 234          | R          |
| 234          | PowerBI    |
| 234          | SQL Server |
| 345          | Python     |
| 345          | Tableau    |

## Example Output:

| candidate_id |
|--------------|
| 123          |

**Solution:**

```
SELECT CANDIDATE_ID FROM CANDIDATES
WHERE SKILL IN ('Python', 'Tableau', 'PostgreSQL')
GROUP BY CANDIDATE_ID HAVING COUNT(SKILL) = 3
ORDER BY CANDIDATE_ID ASC;
```

(2.)

Assume you're given two tables containing data about Facebook Pages and their respective likes (as in "Like a Facebook Page").

Write a query to return the IDs of the Facebook pages that have zero likes. The output should be sorted in ascending order based on the page IDs.

**pages Table:**

| Column Name | Type    |
|-------------|---------|
| page_id     | integer |
| page_name   | varchar |

**pages Example Input:**

| page_id | page_name              |
|---------|------------------------|
| 20001   | SQL Solutions          |
| 20045   | Brain Exercises        |
| 20701   | Tips for Data Analysts |

**page\_likes Table:**

| Column Name | Type     |
|-------------|----------|
| user_id     | integer  |
| page_id     | integer  |
| liked_date  | datetime |

**page\_likes Example Input:**

| user_id | page_id | liked_date          |
|---------|---------|---------------------|
| 111     | 20001   | 04/08/2022 00:00:00 |
| 121     | 20045   | 03/12/2022 00:00:00 |
| 156     | 20001   | 07/25/2022 00:00:00 |

**Example Output:**

| page_id |
|---------|
| 20701   |

**Solution :-**

```
SELECT P.PAGE_ID FROM PAGES AS P  
LEFT JOIN PAGE_LIKES AS L  
ON P.PAGE_ID = L.PAGE_ID  
WHERE L.USER_ID IS NULL  
ORDER BY P.PAGE_ID ASC;
```

**(3.)**

Tesla is investigating production bottlenecks and they need your help to extract the relevant data. Write a query to determine which parts have begun the assembly process but are not yet finished.

Assumptions:

- **parts\_assembly** table contains all parts currently in production, each at varying stages of the assembly process.
- An unfinished part is one that lacks a **finish\_date**.

This question is straightforward, so let's approach it with simplicity in both thinking and solution.

*Effective April 11th 2023, the problem statement and assumptions were updated to enhance clarity.*

### parts\_assembly Table

| Column Name   | Type     |
|---------------|----------|
| part          | string   |
| finish_date   | datetime |
| assembly_step | integer  |

### parts\_assembly Example Input

| part    | finish_date         | assembly_step |
|---------|---------------------|---------------|
| battery | 01/22/2022 00:00:00 | 1             |
| battery | 02/22/2022 00:00:00 | 2             |
| battery | 03/22/2022 00:00:00 | 3             |
| bumper  | 01/22/2022 00:00:00 | 1             |
| bumper  | 02/22/2022 00:00:00 | 2             |
| bumper  |                     | 3             |
| bumper  |                     | 4             |

### Example Output

| part   | assembly_step |
|--------|---------------|
| bumper | 3             |
| bumper | 4             |

### Solution :-

```
SELECT PART, ASSEMBLY_STEP
FROM PARTS_ASSEMBLY
WHERE FINISH_DATE IS NULL;
```